

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Branch : Biomedical Engineering Semester : VI****Subject Code & Name: BTBMC601 & Biomedical Control System Course: T.Y. B.Tech****Max Marks: 60****Date:18/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is an example of an open-loop control system? a. Washing Machine b. Automatic Toaster c. Traffic Light System d. Human Respiratory System	Remember	1
2	What is the purpose of feedback in a control system? a. To increase stability b. To reduce error c. To maintain system performance d. All of the above	Understand	1
3	Which method is used for block diagram simplification? a. Signal flow graphs b. Routh-Hurwitz c. Bode plots d. Nyquist Criterion	Understand	1
4	In time-domain analysis, what is the significance of the steady-state error? a. It measures system stability b. It measures system accuracy c. It indicates oscillatory behavior d. It shows damping ratio	Analyze	1
5	Which test signal is not commonly used in time-domain analysis? a. Step b. Impulse c. Sinusoidal d. Ramp	Understand	1
6	What does a root locus diagram represent? a. System b. Variations of c. Stability d. Frequency	Understand	1

	response over time	poles and zeros with parameter changes	margins	response		
7	What type of model is used to represent physiological systems?				Understand	1
	a. Frequency domain	b. Differential equation-based	c. Statistical	d. Empirical		
8	Which factor primarily governs human thermal regulation?				Remember	1
	a. Heat production	b. Heat loss	c. Homeostasis	d. All of the above		
9	Which of the following is not a frequency domain analysis tool?				Understand	1
	a. Polar plot	b. Root locus	c. Bode plot	d. Nyquist plot		
10	How is gain margin determined from a Bode plot?				Analyze	1
	a. At phase crossover frequency	b. At gain crossover frequency	c. At 0 dB	d. At -180 degrees		
11	What is the primary purpose of the Nyquist plot?				Understand	1
	a. To calculate time constant	b. To determine stability	c. To simplify block diagrams	d. To find steady-state error		
12	Which system is an example of a skeletal muscle servo mechanism?				Remember	1
	a. Cardiac cycle	b. Respiratory cycle	c. Stretch reflex	d. Thermoregulation		
Q. 2	Solve the following.					12
A)	Define and differentiate between open-loop and closed-loop control systems with examples.				Understand	6
B)	Derive the transfer function for a given block diagram using block diagram reduction techniques				Apply	6

Q.3	Solve the following.		12
A)	Explain the Routh-Hurwitz stability criterion with an example	Understand	6
B)	Analyze the time-domain response of a second-order system for a step input and discuss the performance specifications	Analyze	6
Q. 4	Solve Any Two of the following.		12
A)	Develop a compartmental model for a simple physiological system. Discuss the assumptions and limitations	Apply	6
B)	Explain the heat transfer mechanisms within the human body and the role of thermoregulation.	Understand	6
C)	Derive and discuss the relation between time-domain and frequency-domain specifications.	Analyze	6
Q.5	Solve Any Two of the following.		12
A)	Sketch a Bode plot for a given transfer function and determine the stability margins. $T(s) = \frac{100(s+1)}{(s+10)(s+100)}$	Apply	6
B)	Use Nyquist stability criterion to assess the stability of a given system. $G(s) = \frac{1}{s^2 + s + 1}$	Analyze	6
C)	Compare and contrast the Polar plot and Nyquist plot for frequency response analysis. Assume suitable data.	Analyze	6

Q. 6	Solve Any Two of the following.		12
A)	Discuss the control mechanisms of the respiratory system with a relevant model.	Understand	6
B)	Explain the cardiovascular control system and its significance in maintaining homeostasis.	Apply	6
C)	Analyze the role of biological receptors in maintaining stability in physiological systems.	Analyze	6
	*** End ***		